

DOUGLAS RIVER, Australia

WMO: 941280

Lat: 13.8344S

Lon: 131.1872E

Elev: 145

StdP: 14.62

Time Zone: 9.50 (AUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	51.0	53.8	24.0	18.2	80.3	28.9	23.0	76.9	20.4	81.2	18.9	80.3	2.6	140	0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	24.8	100.5	72.0	98.8	72.7	97.3	73.4	85.0	90.1	81.7	89.9	80.6	89.3	7.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.8	177.6	88.9	79.7	155.0	85.3	78.5	148.6	83.7	49.6	90.1	45.6	90.2	44.4	89.5	97.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.5	15.2	13.3	DB	45.2	104.1	2.8	1.7	43.2	105.3	41.6	106.3	40.0	107.2	38.0	108.4
			WB	42.0	89.6	2.0	6.1	40.6	94.0	39.5	97.6	38.4	101.1	37.0	105.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	80.6	83.1	82.9	83.2	81.7	77.5	73.0	72.9	75.2	82.1	85.8	85.6	84.6	(6)
(7)		DBStd	5.63	2.61	2.45	2.04	2.74	4.26	4.86	4.21	4.00	3.51	2.75	2.55	2.66	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	5	0	0	0	0	0	3	2	0	0	0	0	0	(9)
(10)		CDD50	11174	1027	920	1028	951	854	690	711	780	962	1111	1069	1071	(10)
(11)		CDD65	5704	562	500	563	501	389	243	248	315	512	646	619	606	(11)
(12)		CDH74	64921	5798	5008	5857	5434	4084	2588	2901	3971	6565	8434	7472	6809	(12)
(13)		CDH80	32071	2392	2011	2626	2740	1965	1079	1270	2022	3816	4913	4012	3225	(13)
(14)	Wind	WSAvg	4.8	4.0	3.8	3.6	4.7	5.8	6.8	6.3	5.4	4.6	4.6	4.0	3.8	(14)
(15)	Precipitation	PrecAvg	51.2	12.4	11.4	7.6	1.8	0.5	0.0	0.0	0.1	0.2	1.5	5.0	10.6	(15)
(16)		PrecMax	71.4	27.7	22.9	22.6	15.0	4.4	0.1	0.5	0.9	2.3	5.9	9.0	23.5	(16)
(17)		PrecMin	36.4	2.7	4.8	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	1.4	3.7	(17)
(18)		PrecStd	9.6	6.0	4.9	5.6	3.1	1.0	0.0	0.1	0.2	0.4	1.3	1.9	4.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.9	96.2	96.6	97.2	95.5	93.4	92.6	96.2	100.4	103.2	102.0	100.1	(19)
(20)		MCWB	78.7	78.9	76.8	73.8	71.2	70.1	67.7	69.3	69.6	69.7	73.3	76.9	(20)	
(21)		2%	DB	94.5	94.3	94.9	95.4	93.6	90.9	90.8	93.7	98.6	101.0	99.7	97.2	(21)
(22)		MCWB	79.5	78.9	76.9	74.1	70.8	69.0	67.0	68.8	69.8	70.9	74.3	78.0	(22)	
(23)		5%	DB	92.8	92.6	93.4	93.8	92.0	88.8	89.1	91.8	97.0	99.3	97.7	95.2	(23)
(24)		MCWB	79.5	79.0	77.2	74.0	70.3	68.2	66.2	66.9	69.5	71.6	75.5	78.3	(24)	
(25)		10%	DB	90.7	90.6	91.7	92.2	90.0	86.6	87.3	89.9	95.5	97.5	95.6	93.2	(25)
(26)		MCWB	79.7	78.9	77.6	73.8	70.0	66.7	65.2	65.7	69.6	72.1	76.5	78.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	88.2	82.3	84.8	85.5	79.2	80.8	76.9	85.7	89.5	88.3	86.4	81.9	(27)
(28)		MCDWB	89.8	90.7	89.7	91.3	86.0	83.0	84.5	88.2	91.1	95.0	92.8	92.1	(28)	
(29)		2%	WB	82.4	80.9	81.2	79.9	77.1	73.4	72.2	73.7	76.5	79.0	81.7	80.9	(29)
(30)		MCDWB	89.8	89.5	89.2	87.8	84.7	83.8	83.4	84.4	86.9	88.3	91.5	91.0	(30)	
(31)		5%	WB	80.8	80.2	80.1	78.3	75.6	71.3	69.8	71.7	75.3	77.4	79.5	80.1	(31)
(32)		MCDWB	88.7	88.3	88.2	86.0	83.4	83.5	82.6	83.7	85.0	86.8	88.9	89.7	(32)	
(33)		10%	WB	79.9	79.5	79.3	77.3	73.9	69.3	67.8	70.0	74.4	76.5	78.5	79.5	(33)
(34)		MCDWB	87.4	87.1	86.9	85.0	82.9	82.0	81.7	82.3	84.4	86.1	87.8	88.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	14.8	14.8	17.4	22.3	25.9	28.0	29.7	31.0	28.7	24.8	21.1	17.2	(35)	
(36)		5% DB	MCDBR	18.4	18.6	21.1	24.9	27.1	28.6	30.5	31.8	29.4	27.6	24.0	20.7	(36)
(37)		MCWBR	6.0	6.0	6.3	7.0	7.6	9.5	10.3	10.6	8.2	7.1	6.0	5.7	(37)	
(38)		5% WB	MCDBR	16.0	15.9	16.9	19.1	20.4	26.0	27.2	28.2	25.2	23.0	20.1	17.8	(38)
(39)		MCWBR	7.0	6.0	6.5	7.2	7.4	11.9	11.4	12.1	8.1	6.8	7.3	5.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.417	0.409	0.395	0.374	0.369	0.343	0.339	0.357	0.424	0.452	0.462	0.427	(40)	
(41)		taud	2.469	2.486	2.523	2.551	2.503	2.524	2.525	2.455	2.304	2.257	2.273	2.425	(41)	
(42)		Ebn at Noon	294	294	292	288	279	282	286	289	279	279	279	291	(42)	
(43)		Edh at Noon	38	37	35	32	31	30	30	35	43	46	46	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1787	1813	1859	1875	1720	1650	1736	1943	2088	2150	2065	1924	(44)	
(45)		RadStd	175	271	214	137	96	72	49	66	108	109	92	155	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+0.59	N/A	N/A	+0.61	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page